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Project:	test_crack_control_02.py
Subject:	
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Eurocode 2

SLS design of rectangular reinforced concrete cross-section for crack control

Description:

SLS design of rectangular reinforced concrete cross-section for crack control (calculation of required reinforcement for given bending moment / axial force, direct calculation of crack width)

According to:

EN 1992-1-1:2004+AC2:2010 Section 7.3.4

Applicable for:

Uniaxial bending. High bond reinforcement bars (ribbed)

Supported National Annexes:

Nationally Defined Parameters (NDPs) automatically filled for supported countries

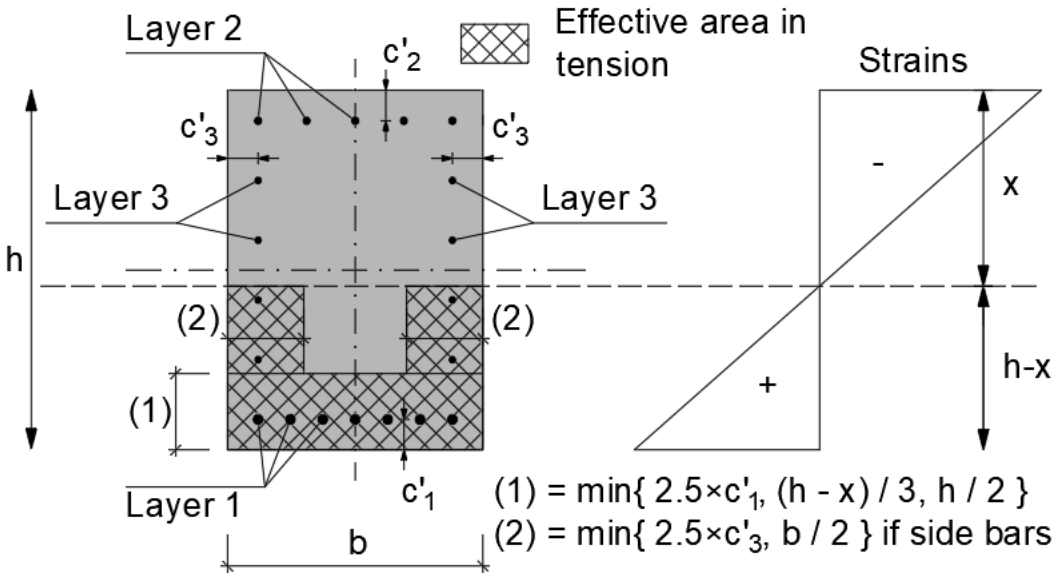
Input

Concrete characteristic strength	f_{ck} =	30	MPa
Normalized maximum compressive stress permitted in the concrete	$\sigma_{c,lim}/f_{ck}$ =	1	
Type of stress-strain law for concrete	=	Parabolic	
Modification factor for modulus of elasticity of concrete	$E_{c,eff}/E_{cm}$ =	1	
Steel characteristic yield strength	f_{yk} =	500	MPa
Maximum tensile stress permitted in the reinforcement	$\sigma_{s,lim}$ =	400	MPa
Height of cross-section	h =	0.4	m
Width of cross-section	b =	1.0	m

Initial reinforcement: When the required reinforcement is requested the initial reinforcement will be modified accordingly.

Number of bars of bottom reinforcement layer 1	$n_{bars,1}$ =	6	
Diameter of bars of bottom reinforcement layer 1	Φ_1 =	20	mm

Distance from centroid of bottom reinforcement layer 1 to bottom concrete edge	c'_1 =	0.068	m
Number of bars of top reinforcement layer 2	$n_{bars,2}$ =	6	
Diameter of bars of top reinforcement layer 2	Φ_2 =	20	mm
Distance from centroid of top reinforcement layer 2 to top concrete edge	c'_2 =	0.068	m
Number of bars of side reinforcement layer 3 per side	$n_{bars,3}$ =	0	
Diameter of bars of side reinforcement layer 3	Φ_3 =	16	mm
Distance from centroid of side reinforcement layer 3 to side concrete edge	c'_3 =	0.07	m
Check crack control for side reinforcement	=	No	
Type of analysis	=	Required reinforcement As	
Design value of the bending moment - SLS combination (positive when tension at bottom side)	M_{Ed} =	0	kNm
Design value of the axial force - SLS combination (compression negative)	N_{Ed} =	253	kN
Variable reinforcement layers to be modified	=	Layers 1 & 2	
Factor dependent on the duration of the load	k_t =	0.4	
Limiting value for the calculated crack width	w_{max} =	0.1	mm
Concrete cover to tensile reinforcement surface	c =	62.33	mm



Rectangular concrete cross-section and effective area in tension for crack control

Nationally Defined Parameters

Coefficients k_3, k_4 for the calculation of maximum crack spacing	= Default (k_3=3.4, k_4=0.425)

Results

Calculated crack width for bottom edge	$w_{k,bot}$ = 0.10 mm
Calculated crack width for top edge	$w_{k,top}$ = 0.10 mm
Calculated steel stress for bottom edge	$\sigma_{s,bot}$ = 64.1 MPa
Calculated steel stress for top edge	$\sigma_{s,top}$ = 64.1 MPa

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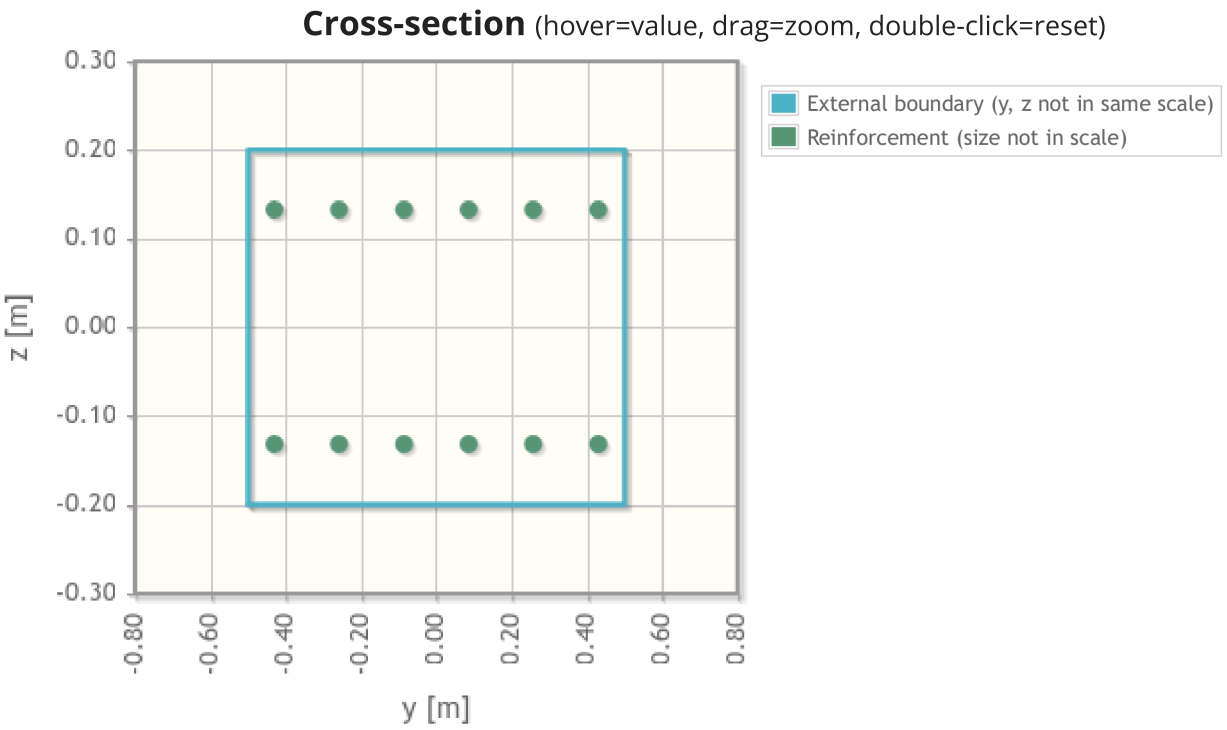
Reinforcement - Layer 1 (bottom)	$A_{s,1}$ = 19.73 cm² (initial×1.047)
Reinforcement - Layer 2 (top)	$A_{s,2}$ = 19.73 cm² (initial×1.047)
Reinforcement - Layer 3 (sides)	$A_{s,3}$ = 0.00 cm²
Reinforcement - Total	$A_{s,tot}$ = 39.47 cm² (0.99%)

Check of total axial force convergence	$\lvert\Delta N\rvert$ = 0.00 N ≤ 1 N ⇒ ok
Check of bending moment convergence	$\lvert\Delta M\rvert$ = 0.00 Nm ≤ 1 Nm ⇒ ok

Notes

1. Positive bending moment produces tension at the bottom side. Negative bending moment produces tension at the top side.
2. The indicated reinforcement layers have been modified by multiplication by the same common factor that is indicated above. The layers in tension are generally more effective. When large compressive axial force is applied increasing the layers in compression may be beneficial.
3. When the required reinforcement is zero for the selected variable layer(s) the stress state that is returned corresponds to equilibrium without the strains of the materials reaching their corresponding limits.

Charts



Tables

Internal equilibrium state of cross-section

	N-M interaction		Lever arm & neutral axis					
#	Axial force N [kN]	Bending moment M [kNm]	Location of neutral axis from centroid y_n [m]	Lever arm of internal forces z [m]	Normalized level arm z/d [-]	Depth of compressive zone x [m]	Normalized depth of compressive zone x/d [-]	Area of compressive zone A_C [m^2]
1	253.00	0.00	Infinity	-	-	0.000	0.000	0.0000

Crack control verification

Edge	Edge in tension	Concrete cover to longitudinal reinforcement surface c [mm]	Depth of effective tensile zone for crack control $h_{c,eff}$ [m]	Area of effective tensile zone for crack control $A_{c,eff}$ [m^2]	Area of reinforcement within the effective tensile zone $A_{s,eff}$ [cm^2]	Effective reinforcement ratio for crack control $\rho_{p,eff}$ [-]	Effect reinforce diamete crack co Φ_{eff} [m
Bottom	yes	62.3	0.170	0.1700	19.73	0.0116	20.0
Top	yes	62.3	0.170	0.1700	19.73	0.0116	20.0

Details

Input Data

- Concrete characteristic strength: f_{ck} = 30 MPa
- Normalized maximum compressive stress permitted in the concrete: $\sigma_{c,lim}/f_{ck}$ = 1
- Type of stress-strain law for concrete: = Parabolic
- Modification factor for modulus of elasticity of concrete: $E_{c,eff}/E_{cm}$ = 1
- Steel characteristic yield strength: f_{yk} = 500 MPa
- Maximum tensile stress permitted in the reinforcement: $\sigma_{s,lim}$ = 400 MPa
- Height of cross-section: h = 0.4 m
- Width of cross-section: b = 1 m
- Number of bars of bottom reinforcement layer 1: $n_{bars,1}$ = 6
- Diameter of bars of bottom reinforcement layer 1: Φ_1 = 20 mm
- Distance from centroid of bottom reinforcement layer 1 to bottom concrete edge: c'_1 = 0.068 m
- Number of bars of top reinforcement layer 2: $n_{bars,2}$ = 6
- Diameter of bars of top reinforcement layer 2: Φ_2 = 20 mm
- Distance from centroid of top reinforcement layer 2 to top concrete edge: c'_2 = 0.068 m
- Number of bars of side reinforcement layer 3 per side: $n_{bars,3}$ = 0
- Diameter of bars of side reinforcement layer 3: Φ_3 = 16 mm
- Distance from centroid of side reinforcement layer 3 to side concrete edge: c'_3 = 0.07 m
- Check crack control for side reinforcement: = No
- Type of analysis: = Required reinforcement As
- Design value of the bending moment - SLS combination (positive when tension at bottom side): M_{Ed} = 0 kNm
- Design value of the axial force - SLS combination (compression negative): N_{Ed} = 253 kN
- Variable reinforcement layers to be modified: = Layers 1 & 2
- Factor dependent on the duration of the load: k_t = 0.4
- Limiting value for the calculated crack width: w_{max} = 0.1 mm
- Concrete cover to tensile reinforcement surface: c = 62.33 mm

Nationally Defined Parameters

- Coefficients k_3, k_4 for the calculation of maximum crack spacing: = Default (k_3=3.4, k_4=0.425)

Properties of rectangular concrete cross-section

The following geometric properties are obtained from basic theory of mechanics. The values of the reinforcement areas, reinforcement ratios and axial resistances of the cross-section correspond to the final reinforcement if different than the initially specified reinforcement.

- Area of gross cross-section: A = 0.4000 m^2
- Second moment of the area of gross cross-section: I = 0.005333 m^4
- Total height of cross-section: h = 0.400 m
- Location of centroid of gross cross-section (from top): z_{cm} = 0.200 m
- Total area of longitudinal reinforcement: $A_{s,tot}$ = 39.47 cm^2
- Geometric reinforcement ratio: $\rho_l = A_{s,tot} / A$ = 0.0099 = 0.99 %

Stress-strain law at the Serviceability Limit State (SLS)

Assumptions

The design SLS bending moment resistance M_{Rd} of reinforced concrete cross-section is obtained similarly to the procedure applicable for ULS that is described in *EN1992-1-1 §6.1*. This analysis is applicable for undisturbed regions of beams, columns, slabs and other similar types of members for which sections remain plane before and after loading. Discontinuity regions are not covered in this calculation, see *EN 1992-1-1 §6.5* instead.

The following assumptions are made in *EN1992-1-1 §6.1(2)P*:

- Plane sections remain plane after loading.
- The strain in bonded reinforcement or bonded prestressing tendons, whether in tension or in compression, is the same as that in the surrounding concrete.
- The tensile strength of the concrete is ignored.
- The initial strain in prestressing tendons is taken into account when assessing the stresses in the tendons.

Parabolic stress-strain relation for concrete

According to the general parabolic stress-strain law, the compressive stress of concrete σ_c is obtained from the

design curves in *EN1992-1-1 §3.1.7* and *EN1992-1-1 Figure 3.3* as a function of the concrete strain ε_c. According to *EN1992-1-1 §2.4.2.4(2)* the material partial factor for concrete in SLS is $\gamma_C = 1.0$.

$\sigma_c = 0$ for tensile strain
$\sigma_c = f_{ck} \cdot [1 - (1 - \varepsilon_c / \varepsilon_{c2})^n]$ for compressive strain $0 \le \varepsilon_c \le \varepsilon_{c2}$
$\sigma_c = f_{ck}$ for compressive strain $\varepsilon_{c2} < \varepsilon_c \le \varepsilon_{cu2}$

The strains ε_{c2}, ε_{cu2} and the exponent n are defined in *EN1992-1-1 Table 3.1* depending on the concrete class. For characteristic strength $f_{ck} = 30.00$ MPa the corresponding values are $\varepsilon_{c2} = 3.50$ ‰, $\varepsilon_{cu2} = 3.50$ ‰ and $n = 2.00$.

Linear stress-strain relation for concrete

As a simplification a linear stress-strain relation can be assumed for concrete in compression. The compressive stress of concrete σ_c is obtained as a linear function of the concrete strain ε_c:

$\sigma_c = 0$ for tensile strain
$\sigma_c = E_{c,eff} \cdot \varepsilon_c$ for compressive strain

Where $E_{c,eff}$ is the effective modulus of elasticity for concrete to be considered in the analysis. In general $E_{c,eff}$ can be specified equal to the secant modulus of elasticity of concrete E_{cm} as defined in *EN1992-1-1 Table 3.1* depending on the concrete class. For characteristic compressive strength $f_{ck} = 30.00$ MPa the corresponding value of the secant modulus of elasticity of concrete is $E_{cm} = 32837$ MPa. The value of $E_{c,eff}$ can take into account more refined values of the modulus of elasticity based on the aggregate type an appropriate value can be provided, e.g. as specified in *EN1992-1-1 §3.1.3(2)* and the National Annex. To take into account the long-term effects of creep the effective value of the modulus of elasticity $E_{c,eff}$ can be defined as specified in *EN1992-1-1 §7.4.3(5)*:

$E_{c,eff} = E_{cm} / [1 + \varphi(\infty, t_0)]$

Stress-strain relation for reinforcement steel

The stress in reinforcement steel σ_s is obtained from the design curves in *EN1992-1-1 §3.2.7* and *EN1992-1-1 Figure 3.8* as a function of the steel strain ε_s. According to *EN1992-1-1 §2.4.2.4(2)* the material partial factor for reinforcement steel in SLS is $\gamma_S = 1.0$.

$\sigma_s = E \cdot \varepsilon_s$ for strains below the design yield point $-\varepsilon_{yk} \le \varepsilon_s \le \varepsilon_{yk}$
$\sigma_s = +f_{yk}$ for tensile strain $\varepsilon_s \ge \varepsilon_{yk}$
$\sigma_s = -f_{yk}$ for compressive strain $\varepsilon_s < -\varepsilon_{yk}$

The characteristic yield strain of the reinforcement steel ε_{yk} is calculated based on the modulus of elasticity specified in *EN1992-1-1 §3.2.7(4)*:

$\varepsilon_{yk} = f_{yk} / E = 500.00$ MPa $/ 200000$ MPa $= 2.50$ ‰

In the SLS in general the tensile stress of the reinforcement steel is limited to a value below the yield stress f_{yk}. For the compressive stress of reinforcement steel values smaller than the yield stress $-f_{yk}$ are not considered e.g if the reinforcement steel yields in compression before concrete reaches its stress limit. This is compatible with the assumption of a horizontal top branch without the need to check the strain limit for the reinforcement steel stress-strain law, that is permitted according to *EN1992-1-1 §3.2.7(2)*.

Permitted range of strain distributions

The permitted range of strain distributions is shown in *EN1992-1-1 Figure 6.1*. The range of permitted strain distributions correspond to linear strain profiles that are defined by the following rules:

- Limitation of tensile strain in reinforcement steel to $\varepsilon_{s,max}$
- Limitation of compressive strain in concrete to $\varepsilon_{c,max}$

The maximum permitted tensile strain in reinforcement $\varepsilon_{s,max}$ corresponds to the maximum permitted tensile stress in the reinforcement $\sigma_{s,max}$ so as to fulfill the crack width verification (see below).

$\varepsilon_{s,max} = \sigma_{s,max} / E$

The maximum permitted compressive strain in concrete $\varepsilon_{c,max}$ corresponds to the maximum permitted compressive stress in the concrete $\sigma_{c,max}$: The value of $\varepsilon_{c,max}$ can be obtained from the value of $\sigma_{c,max}$ by inverting the applicable stress-strain function of concrete.

For a cross-section subjected to bending moment inducing tension at the bottom edge and if d is the static depth from most distant reinforcement to the upper concrete edge then the following strain distribution ranges

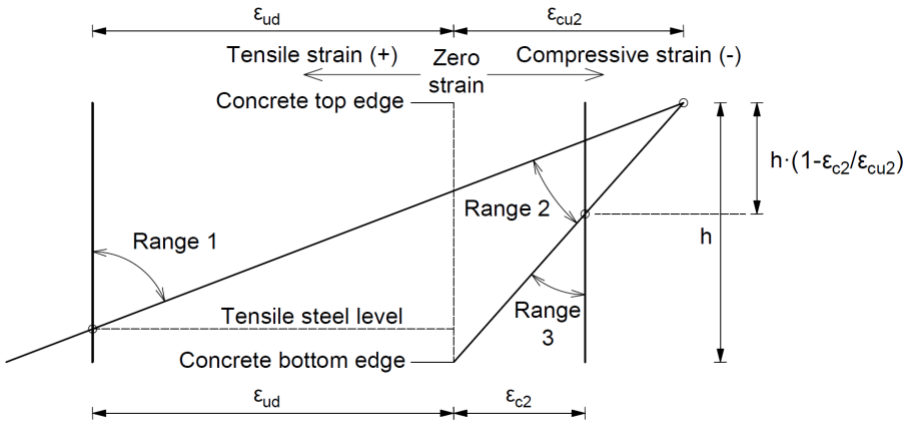
are established:

- Range 1: Constant tensile strain $\varepsilon_{s,max}$ at reinforcement level (z=*d*). Varying strain at top edge (z=0) from $\varepsilon_{s,max}$ to $-\varepsilon_{c,max}$.
- Range 2: Constant compressive strain $-\varepsilon_{c,max}$ at top edge (z=0). Varying strain at lower part from $\varepsilon_{s,max}$ at level z=*d* to 0 at level z=*h*.
- Range 3: Constant compressive strain $-\varepsilon_{c,max}$ at top edge (z=0). Varying strain at bottom edge (z=*h*) from 0 to $-\varepsilon_{c,max}$.

Where *h* is the full height of the cross-section and *d* is the static depth from most distant reinforcement to the upper concrete edge.

For a cross-section subjected to bending moment inducing tension at the top edge the aforementioned ranges are similar with the top and bottom edges swapped.

The permitted range of strain distributions are shown in the following figure (reproduced from EN 1992-1-1 Figure 6.1).



Permitted range of strain distributions for ULS design of concrete cross-sections (according to EN 1992-1-1 Figure 6.1)

The depth of the compressive zone *x* is determined from depth where the strain profile becomes zero.

$$x = h \cdot \varepsilon_{c,top} / (\varepsilon_{c,top} - \varepsilon_{c,bot})$$

The curvature of the cross-section φ corresponding to the strain profile is:

$$\varphi = (\varepsilon_{c,bot} - \varepsilon_{c,top}) / h$$

where $\varepsilon_{c,top}$, $\varepsilon_{c,bot}$ are the strains of the concrete at the top edge and bottom edge respectively (assuming that the bending moment induces tension on the bottom side).

Calculation of internal forces and moments

Discretization of concrete and steel areas

The concrete part of the cross-section is discretized in 200 horizontal strips with each strip having height dh = h / 200, where h the full height of the examined cross-section. The width of the i-th strip b_i is established from geometry of the outer boundary and the inner boundaries of the cross-section. The area of the strip A_i is equal to $A_i = b_i \cdot dh$. The area of reinforcement bars within the concrete strip height is not subtracted from the area of the concrete strip in order to simplify calculations. This leads to a small discrepancy that is generally acceptable. For each strip the strain ε_i is assumed constant and taken equal to the corresponding strain at mid-height of the strip, as established from the examined linear strain profile.

The rebars are considered as concentrated point forces at their center. The area of each rebar is $A_i = \pi \Phi_i^2/4$. When non-integer number of bars is specified the next integer number of bars is applied with a suitably reduced area for each rebar.

The force F_i of each component of the cross-section (concrete strip or rebar) is $F_i = A_i \cdot \sigma_i$, where σ_i the stress at the center of the component that corresponds to the relevant strain ε_i. The total axial force N is the summation of all force components F_i.

The moment M_i of each component of the cross-section (concrete strip or rebar) is $M_i = F_i \cdot (z_i - z_{cm})$, where z_i is the location of the center of the component (measured from top) and z_{cm} the location of the centroid of the

gross cross-section. The total bending moment M is the summation of all moment components M_i.

Pair of internal forces

The total force of the cross-section is divided in a tensile component F_T (from rebars in tension) and a compressive component F_C (from concrete in compression and rebars in compression). The internal force components are located at depths z_T and z_C respectively. The internal moment components are $M_T = F_T \cdot (z_T - z_{cm})$ and $M_C = F_C \cdot (z_C - z_{cm})$. The lever arm z of the internal forces is:

$$z = z_T - z_C$$

Calculation of required reinforcement for given for given bending moment M_{Ed} and axial force N_{Ed}

The required reinforcement area A_s is calculated in order to obtain a bending moment resistance M_{Rd} at least equal to the bending moment M_{Ed} for given and axial force N_{Ed}. A common factor is calculated that multiplies the reinforcement areas of the selected reinforcement layers. The rest of the reinforcement layers are not affected. The value of the multiplication factor is obtained by a root finding algorithm based on the ITP method so that the bending moment resistance M_{Rd} becomes equal to the specified bending moment M_{Ed} for given and axial force N_{Ed}.

It is noted that when two or more layers are modified to obtain the required reinforcement the resulting reinforcement distribution may not be the optimal (i.e. corresponding to the least amount of reinforcement). This is because the variable reinforcement layers are all multiplied by a common factor.

Equilibrium of cross-section

The equilibrium of the cross-section is obtained iteratively by changing the common reinforcement factor and the limit of maximum steel stress $\sigma_{s,max}$ so that a) the equilibrium of internal forces of the cross-section correspond to the given external loads (bending moment M_{Ed} and axial force N_{Ed}), b) the calculated crack width w_k satisfies the crack control verification $w_k \leq w_{max} = 0.10$ mm, and c) the most compressive concrete stress $\sigma_{c,min}$ satisfies the required limit $|\sigma_{c,min}| \leq 1.000 \cdot f_{ck}$. For the examined calculation the value of the steel stress which is obtained iteratively is $\sigma_{s,max} = 64.1$ MPa. The equilibrium of the cross-section is obtained as follows:

- Strain at bottom edge: $\varepsilon_{bottom} = 0.321$ ‰
- Strain at top edge: $\varepsilon_{top} = 0.321$ ‰
- Depth of compressive zone: $x = 0.0000$ m
- Most compressive concrete stress: $\sigma_{c,min} = 0.0$ MPa
- Least compressive concrete stress: $\sigma_{c,max} = 0.0$ MPa
- Most tensile steel stress: $\sigma_{s,max} = 64.1$ MPa
- Least tensile steel stress: $\sigma_{s,min} = 64.1$ MPa

Auxiliary quantities

- Coefficient k_t is a factor dependent on the duration of the load which takes values $k_t = 0.6$ for short-term loading and $k_t = 0.4$ for lond-term loading, with intermediate values possible for a combination of load durations. In this calculation the considered value is $k_t = 0.400$.
- The modulus of elasticity of reinforcement steel is $E_s = 200000$ MPa as specified in *EN 1992-1-1 §3.2.7(4)*.
- The secant modulus of elasticity of concrete E_{cm} is considered according to *EN 1992-1-1 Table 3.1*. The value corresponding to the characteristic compressive strength $f_{ck} = 30.00$ MPa is $E_{cm} = 32837$ MPa.
- An effective value of the modulus of elasticity of concrete can be considered to take into account the effect of creep, aggregate type etc. In this calculation the value of the effective modulus of elasticity of concrete is considered equal to $E_{c,eff} = 1.000 \cdot E_{cm} = 1.000 \cdot 32837$ MPa $= 32837$ MPa.
- The ratio of the elastic modulus of reinforcement to the elastic modulus of concrete is $\alpha_e = E_s / E_{c,eff} = 200000$ MPa / 32837 MPa $= 6.091$
- In the examined calculation the general parabolic stress-strain law is considered for concrete in compression according to *EN1992-1-1 §3.1.7*. The value of the effective elastic modulus $E_{c,eff} = 32837$ MPa as mentioned above is considered only for for the estimation of the ratio of the elastic modulus of reinforcement to the elastic modulus of concrete α_e. The parabolic stress-strain law for concrete in compression is not affected by $E_{c,eff}$.
- The mean value of the tensile strength of the concrete $f_{ct,eff}$ effective at the time when the cracks may first be expected to occur is considered equal to the mean tensile strength of concrete $f_{ctm} = 2.90$ MPa assuming conservatively that cracks have not formed earlier than 28 days. According to *EN 1992-1-1 Table 3.1* the value corresponding to the characteristic compressive strength $f_{ck} = 30.00$ MPa is $f_{ct,eff} = f_{ctm} = 2.90$ MPa.

Tensile zone effective for crack control

The height of the tensile zone that is effective for crack control $h_{c,eff}$ is calculated in general according to *EN 1992-1-1 §7.3.2(3)*:

$$h_{c,eff} = \min \{ 2.5 \cdot (h - d), (h - x) / 3, h / 2 \}$$

where $(h - d)$ is the distance from the centroid of the main tensile reinforcement to the nearest concrete surface (denoted in this calculation as c'_1 for the bottom surface and c'_2 for the top surface).

When side reinforcement is present the effective tensile zone for side reinforcement at the sides of the cross-section is determined as follows:

- Vertically the part of the height of the tension zone $(h - x)$ that does not belong to the effective tensile zones of the bottom surface and top surface.
- Horizontally for a distance $2.5 \cdot c'_3$ from the side surface but limited by half of the width of the cross-section $b / 2$

The aforementioned rules are presented graphically in the figure above.

Calculation of crack width for bottom edge

Effective reinforcement ratio

The height of the tensile zone effective for crack control $h_{c,eff}$ for the bottom edge is calculated according to *EN 1992-1-1 §7.3.2(3)*:

$$h_{c,eff} = \min \{ 2.5 \cdot (h - d), (h - x) / 3, h / 2 \} = \min \{ 2.5 \cdot (0.068 \text{ m}), (0.400 \text{ m} - 0.0000 \text{ m}) / 3, 0.400 \text{ m} / 2 \} = 0.170 \text{ m}$$

The corresponding area of the tensile zone effective for crack control (taking into account the contribution of the side reinforcement when present) is calculated according to the rules presented above as $A_{c,eff} = 0.1700 \text{ m}^2$. The area of the reinforcement contained within the aforementioned effective area is $A_{s,eff} = 19.73 \text{ cm}^2$. Therefore the effective reinforcement ratio for crack control verification is:

$$\rho_{p,eff} = A_{s,eff} / A_{c,eff} = 19.73 \text{ cm}^2 / 0.1700 \text{ m}^2 = 0.0116$$

The equivalent diameter for the reinforcement contained in the effective tensile zone is calculated according to the rule described in *EN 1992-1-1 eq. (7.12)*:

$$\Phi_{eff} = \Sigma \Phi_i^2 / \Sigma \Phi_i = 20.00 \text{ mm}$$

Maximum crack spacing

The maximum crack spacing $s_{r,max}$ is specified in *EN 1992-1-1 §7.3.4(3)*:

$$s_{r,max} = k_3 \cdot c + k_1 \cdot k_2 \cdot k_4 \cdot \Phi_{eff} / \rho_{p,eff}$$

The coefficient k_1 takes account of the bond properties of the bonded reinforcement. For high bond bars according to *EN 1992-1-1 §7.3.4(3)* the corresponding value is $k_1 = 0.8$.

The coefficient k_2 takes into account the ratio of strains at the boundaries of the considered section. For the examined case the strains at the boundaries of the full cross-section are considered: $\varepsilon_1 = 0.321$ ‰ and $\varepsilon_2 = 0.321$ ‰, where ε_1 is the most tensile value.

For the case of tension within the examined boundaries (i.e. ε_1 and ε_2 both positive) the corresponding value is:

$$k_2 = (\varepsilon_1 + \varepsilon_2) / (2 \cdot \varepsilon_1) = (0.321 \text{ ‰} + 0.321 \text{ ‰}) / (2 \cdot 0.321 \text{ ‰}) = 1.000$$

According to *EN 1992-1-1 §7.3.4(3)* the values of the coefficients k_3, k_4 for the calculation of the maximum crack spacing $s_{r,max}$ are considered equal to $k_3 = 3.400$ and $k_4 = 0.425$.

By substituting in the previous equation the maximum crack spacing is:

$$s_{r,max} = 3.400 \cdot 62.3 \text{ mm} + 0.8 \cdot 1.000 \cdot 0.425 \cdot 20.00 \text{ mm} / 0.0116 = 797.7 \text{ mm}$$

It is also verified that the calculated maximum crack spacing does not exceeded the upper bound that is defined in *EN 1992-1-1 eq. (7.14)*:

$$s_{r,max} = 1.3 \cdot (h - x) = 1.3 \cdot (0.400 \text{ m} - 0.0000 \text{ m}) = 520.0 \text{ mm}$$

Therefore the calculated maximum crack spacing is $s_{r,max} = 520.0$ mm

Difference of mean strains for reinforcement and concrete

The difference of the mean strains ε_{sm} - ε_{cm} of reinforcement ε_{sm} and concrete ε_{sm} is obtained according to *EN 1992-1-1 §7.3.4(2)*:

$$\varepsilon_{sm} - \varepsilon_{cm} = \sigma_s / E_s \cdot \max \{ 0.6, 1 - k_t \cdot f_{ct,eff} / \sigma_s \cdot (1 / \rho_{p,eff} + \alpha_e) \}$$

Therefore:

$$\varepsilon_{sm} - \varepsilon_{cm} = \sigma_s / E_s \cdot \max \{ 0.6, 1 - 0.400 \cdot 2.90 \text{ MPa} / 64.1 \text{ MPa} \cdot (1 / 0.0116 + 6.091) \} = \max \{ 0.6, -0.667 \} \cdot \sigma_s / E_s = 0.600 \cdot 64.1 \text{ MPa} / 200000 \text{ MPa} = 0.192 ‰$$

Calculation of crack width

The value of crack width w_k is calculated according to *EN 1992-1-1 §7.3.4(1)*:

$$w_k = (\varepsilon_{sm} - \varepsilon_{cm}) \cdot s_{r,max} = (0.192 ‰) \cdot 520.0 \text{ mm} = 0.10 \text{ mm}$$

The calculated value for the crack width w_k = 0.10 mm does not exceed the limit w_{max} = 0.10 mm. Therefore the crack control verification is satisfied for the bottom edge of the cross-section.

Calculation of crack width for top edge

Effective reinforcement ratio

The height of the tensile zone effective for crack control $h_{c,eff}$ for the top edge is calculated according to *EN 1992-1-1 §7.3.2(3)*:

0.1333 m

$$h_{c,eff} = \min \{ 2.5 \cdot (h - d), (h - x) / 3, h / 2 \} = \min \{ 2.5 \cdot (0.068 \text{ m}), (0.400 \text{ m} - 0.0000 \text{ m}) / 3, 0.400 \text{ m} / 2 \} = 0.170 \text{ m}$$

The corresponding area of the tensile zone effective for crack control (taking into account the contribution of the side reinforcement when present) is calculated according to the rules presented above as $A_{c,eff}$ = 0.1700 m^2. The area of the reinforcement contained within the aforementioned effective area is $A_{s,eff}$ = 19.73 cm^2. Therefore the effective reinforcement ratio for crack control verification is:

0.0147975 m

$$\rho_{p,eff} = A_{s,eff} / A_{c,eff} = 19.73 \text{ cm}^2 / 0.1700 \text{ m}^2 = 0.0116$$

The equivalent diameter for the reinforcement contained in the effective tensile zone is calculated according to the rule described in *EN 1992-1-1 eq. (7.12)*:

$$\Phi_{eff} = \Sigma \Phi_i^2 / \Sigma \Phi_i = 20.00 \text{ mm}$$

Maximum crack spacing

The maximum crack spacing $s_{r,max}$

$$s_{r,max} = k_3 \cdot c + k_1 \cdot k_2 \cdot k_4 \cdot \Phi_{eff} / \rho_{p,eff}$$

The coefficient k_1 takes account of the bond properties of the bonded reinforcement. For high bond bars according to *EN 1992-1-1 §7.3.4(3)* the corresponding value is k_1 = 0.8.

The coefficient k_2 takes into account the ratio of strains at the boundaries of the considered section. For the examined case the strains at the boundaries of the full cross-section are considered: ε_1 = 0.321 ‰ and ε_2 = 0.321 ‰, where ε_1 is the most tensile value.

For the case of tension within the examined boundaries (i.e. ε_1 and ε_2 both positive) the corresponding value is:

$$k_2 = (\varepsilon_1 + \varepsilon_2) / (2 \cdot \varepsilon_1) = (0.321 ‰ + 0.321 ‰) / (2 \cdot 0.321 ‰) = 1.000$$

According to *EN 1992-1-1 §7.3.4(3)* the values of the coefficients k_3, k_4 for the calculation of the maximum crack spacing $s_{r,max}$ are considered equal to k_3 = 3.400 and k_4 = 0.425.

By substituting in the previous equation the maximum crack spacing is:

0.1333 m

0.67136 m

$$s_{r,max} = 3.400 \cdot 62.3 \text{ mm} + 0.8 \cdot 1.000 \cdot 0.425 \cdot 20.00 \text{ mm} / 0.0116 = 797.7 \text{ mm}$$

It is also verified that the calculated maximum crack spacing does not exceeded the upper bound that is defined in *EN 1992-1-1 eq. (7.14)*:

~~$s_{r,max} = 1.3 \cdot (h - x) = 1.3 \cdot (0.400 \text{ m} - 0.0000 \text{ m}) = 520.0 \text{ mm}$~~

This upper bound applies only when the spacing of the bonded reinforcement exceeds 5(c+¢/2) or where there is no bonded reinforcement within the tension zone. This is not the case.

Therefore the calculated maximum crack spacing is $s_{r,max}$ = ~~520.0 mm~~ 0.67136 m

Difference of mean strains for reinforcement and concrete

The difference of the mean strains ε_{sm} - ε_{cm} of reinforcement ε_{sm} and concrete ε_{sm} is obtained according to *EN 1992-1-1 §7.3.4(2)*:

$$\varepsilon_{sm} - \varepsilon_{cm} = \sigma_s / E_s \cdot \max \{ 0.6, 1 - k_t \cdot f_{ct,eff} / \sigma_s \cdot (1 / \rho_{p,eff} + \alpha_e) \}$$

Therefore:

$$\varepsilon_{sm} - \varepsilon_{cm} = \sigma_s / E_s \cdot \max \{ 0.6, 1 - 0.400 \cdot 2.90 \text{ MPa} / 64.1 \text{ MPa} \cdot (1 / 0.0116 + 6.091) \} = \max \{ 0.6, -0.667 \} \cdot \sigma_s / E_s = 0.600 \cdot 64.1 \text{ MPa} / 200000 \text{ MPa} = 0.192 ‰$$

Calculation of crack width

The value of crack width w_k is calculated according to *EN 1992-1-1 §7.3.4(1)*:

$$w_k = (\varepsilon_{sm} - \varepsilon_{cm}) \cdot s_{r,max} = (0.192 ‰) \cdot \text{~~520.0 mm~~ 0.67136 m} = \text{~~0.10 mm~~ 0.1289 mm}$$

The calculated value for the crack width w_k = 0.10 mm does not exceed the limit w_{max} = 0.10 mm. Therefore the crack control verification is satisfied for the top edge of the cross-section.

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